Mini Projet

Docker Virtualisation des machines

travail réalisé par eya chaabene

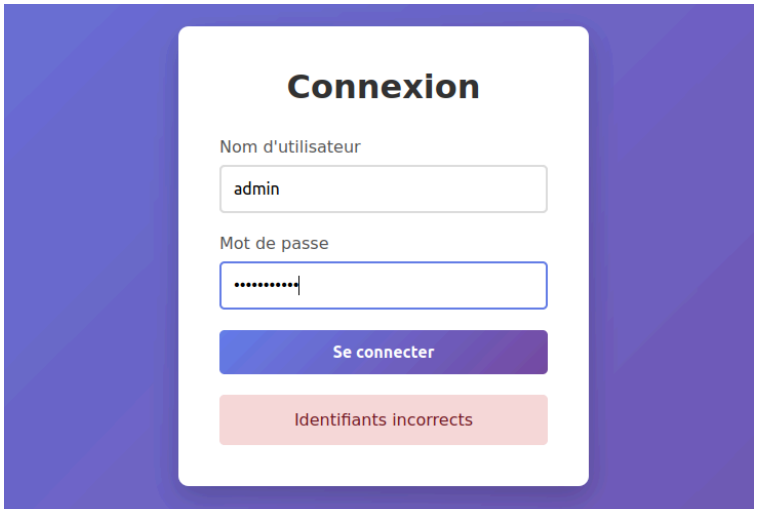
Méthode 1 : Dockerfile (déploiement manuel)

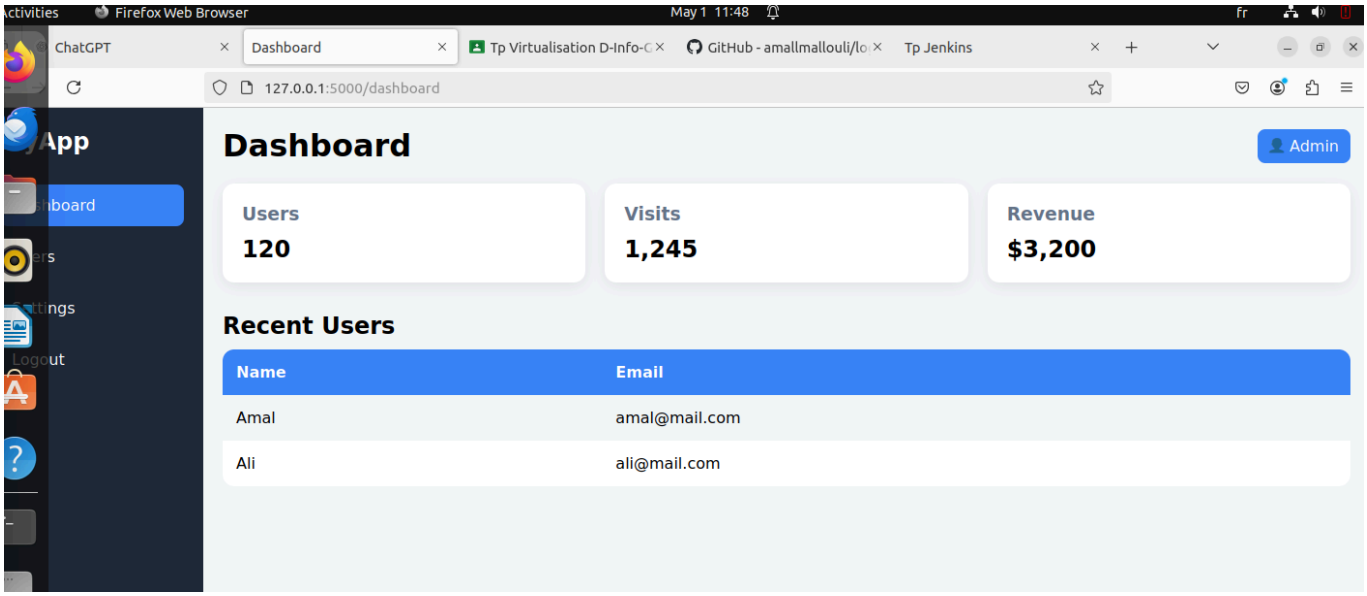
```
vboxuser@virtualisation:~$ git clone https://github.com/amallmallouli/login-web-app
Cloning into 'login-web-app'...
remote: Enumerating objects: 68, done.
remote: Counting objects: 100% (68/68), done.
remote: Compressing objects: 100% (63/63), done.
remote: Total 68 (delta 25), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (68/68), 19.65 KiB | 745.00 KiB/s, done.
Resolving deltas: 100% (25/25), done.
```

```
Activities   Terminal                                   May 1  11:44
vboxuser@virtualisation: ~/login-web-app
  GNU nano 6.2                                   Dockerfile *
# Choisir l'image de base
FROM python:3.11-slim

WORKDIR /app

# Copier les fichiers de requirements
COPY requirements.txt .

# Installer les dépendances
RUN pip install --no-cache-dir -r requirements.txt

# Copier le fichier de l'application
COPY app.py .
COPY templates/ templates/
# Exposer le port
EXPOSE 5000

# Commande de demarrage
CMD ["python", "app.py"]
```

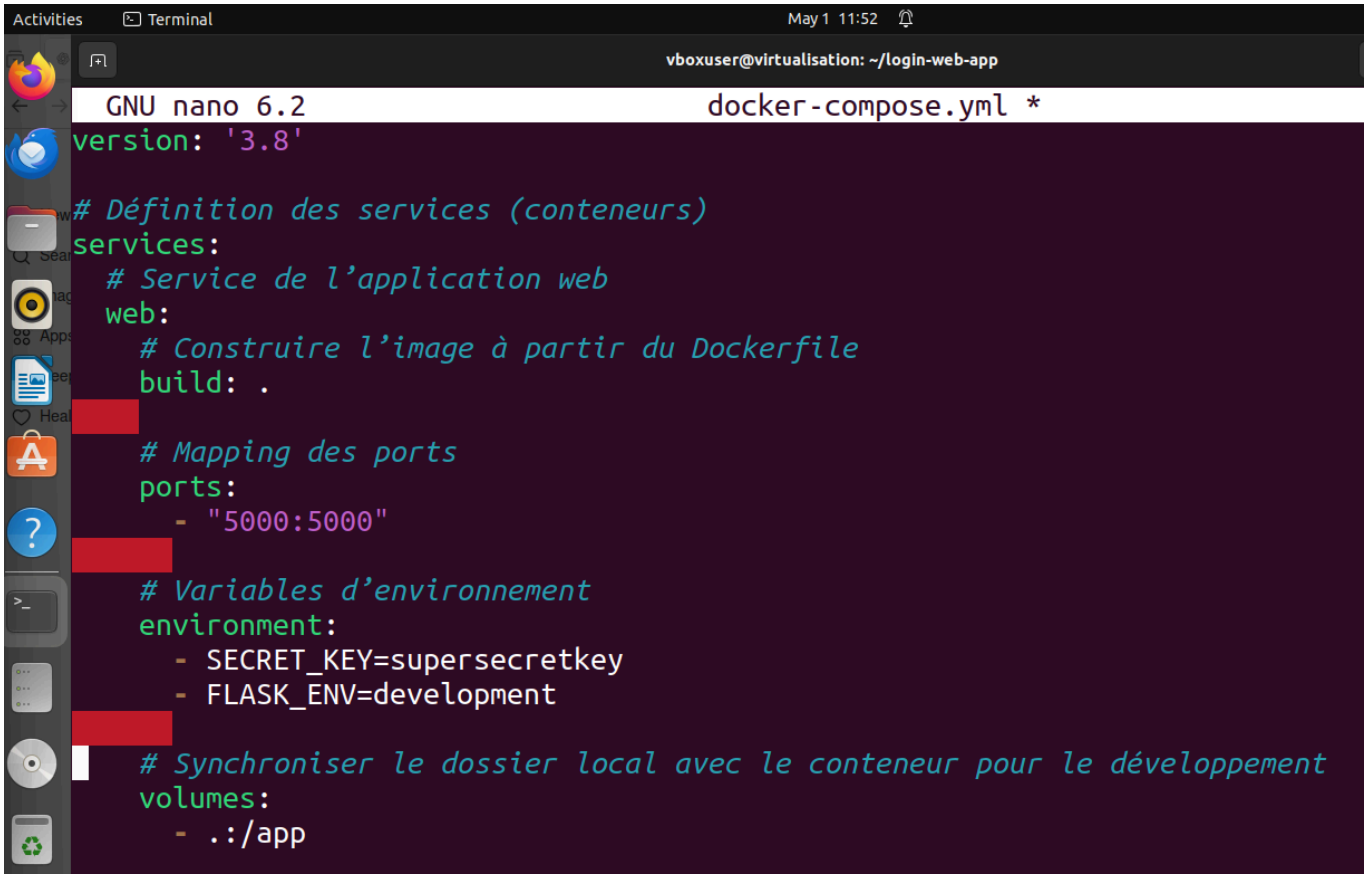
```
vboxuser@virtualisation:~/login-web-app$ nano Dockerfile
vboxuser@virtualisation:~/login-web-app$ docker build -t mylogin .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  98.82kB
Step 1/8 : FROM python:3.11-slim
 ---> 1eee6fcc4d86
Step 2/8 : WORKDIR /app
 ---> Using cache
 ---> 518ac1937101
Step 3/8 : COPY requirements.txt .
 ---> 36f2606e5266
Step 4/8 : RUN pip install --no-cache-dir -r requirements.txt
 ---> Running in 3d3db8dcf706
Collecting Flask==3.0.0 (from -r requirements.txt (line 1))
  Downloading flask-3.0.0-py3-none-any.whl.metadata (3.6 kB)
Collecting Werkzeug==3.0.1 (from -r requirements.txt (line 2))
  Downloading werkzeug-3.0.1-py3-none-any.whl.metadata (4.1 kB)
Collecting Jinja2>=3.1.2 (from Flask==3.0.0->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.1.2 (from Flask==3.0.0->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
```

```
vboxuser@virtualisation:~/login-web-app$ docker run --name login -p 5000:5000 mylogin
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a
duction WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 552-956-671
172.17.0.1 - - [01/May/2026 10:45:54] "GET / HTTP/1.1" 200 -
```

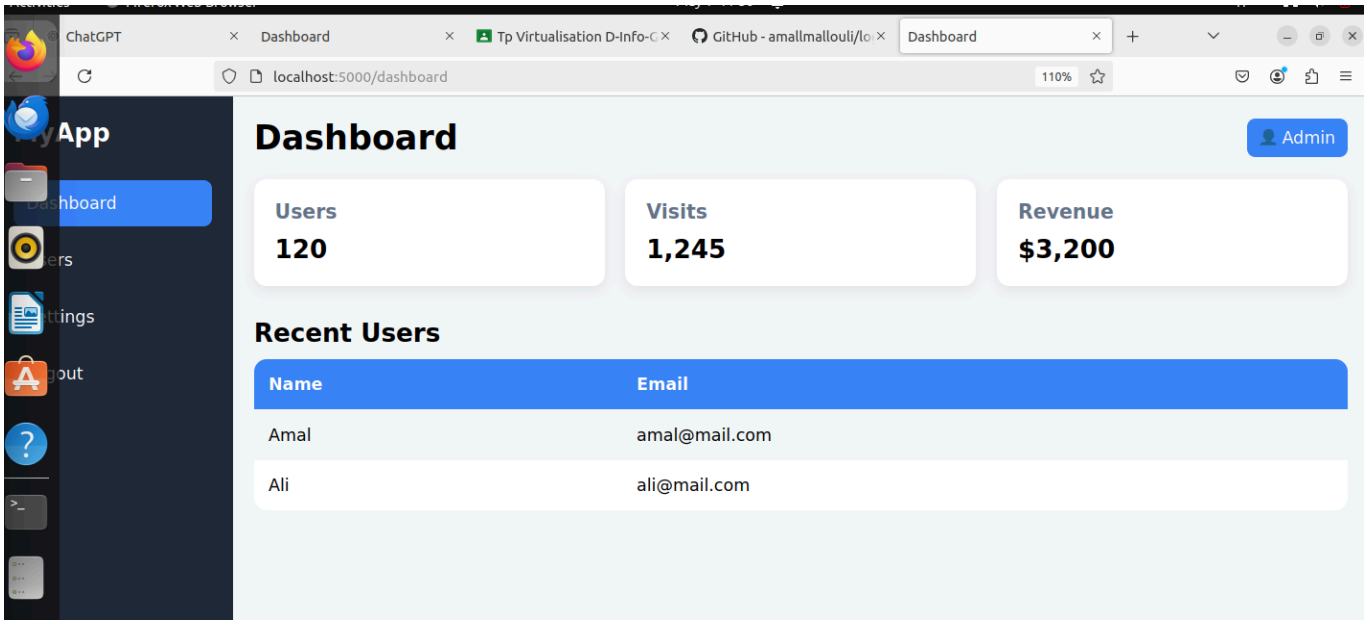




Méthode 2 : Docker Compose (déploiement automatisé)

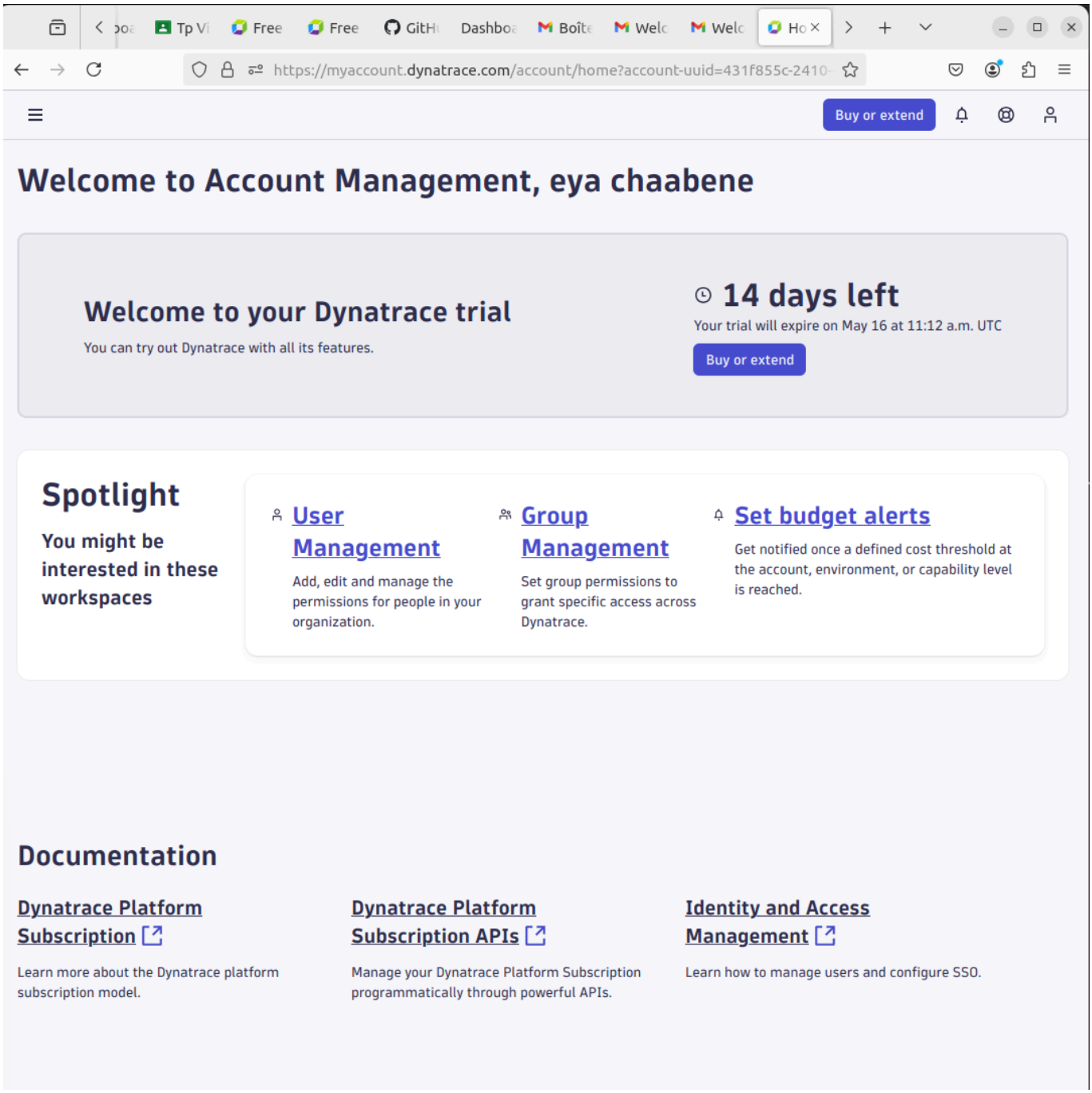


```
vboxuser@virtualisation:~/login-web-app$ nano docker-compose.yml
vboxuser@virtualisation:~/login-web-app$ docker-compose up -d
Creating network "login-web-app_default" with the default driver
Building web
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  98.82kB
Step 1/8 : FROM python:3.11-slim
 ---> 1eee6fcc4d86
Step 2/8 : WORKDIR /app
 ---> Using cache
 ---> 518ac1937101
Step 3/8 : COPY requirements.txt .
 ---> Using cache
 ---> 36f2606e5266
Step 4/8 : RUN pip install --no-cache-dir -r requirements.txt
 ---> Using cache
 ---> 65ca285974e6
Step 5/8 : COPY app.py .
 ---> Using cache
 ---> 2b8c097d9828
Step 6/8 : COPY templates/ templates/
 ---> Using cache
 ---> f6787c275b6c
Step 7/8 : EXPOSE 5000
 ---> Using cache
 ---> d510fd61323a
Step 8/8 : CMD ["python", "app.py"]
 ---> Using cache
 ---> 92da241e3d6e
Successfully built 92da241e3d6e
Successfully tagged login-web-app_web:latest
WARNING: Image for service web was built because it did not already exist. To rebuild this
 image you must use `docker-compose build` or `docker-compose up --build`.
Creating login-web-app_web_1 ... done
vboxuser@virtualisation:~/login-web-app$
```

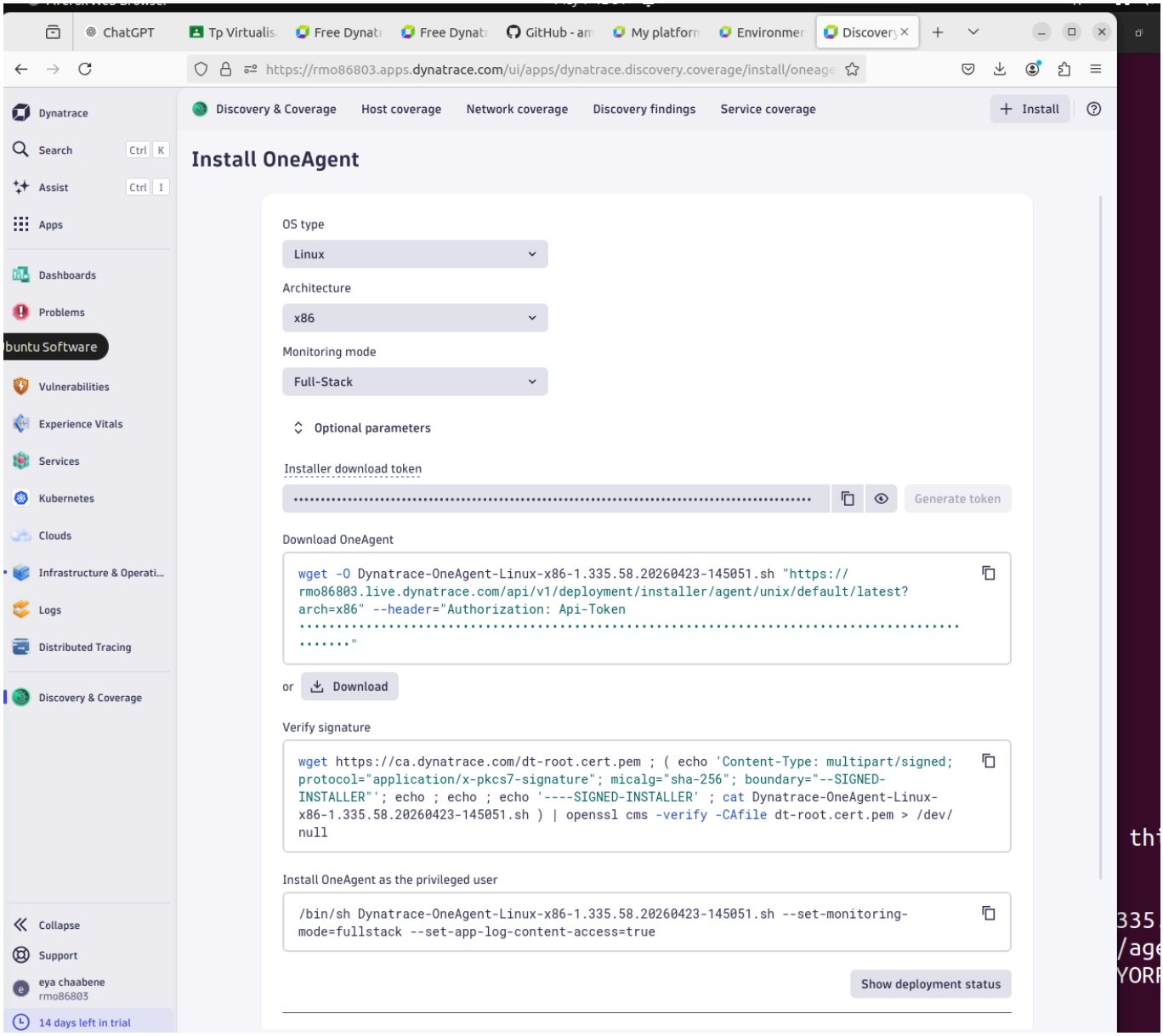


Supervision avec Dynatrace

Étape 1 : Créer ton compte Dynatrace (Free Trial)



Étape 2 : Installer le OneAgent Dynatrace




```
vboxuser@virtualisation:~/login-web-app$ sudo wget -O Dynatrace-OneAgent-Linux-x86-1.335.58.20260423-145051.sh "https://rmo86803.live.dynatrace.com/api/v1/deployment/installer/agent/unix/default/latest?arch=x86" --header="Authorization: Api-Token dt0c01.5J2O7EHWQ2EYORRXEB4L7LOZ.KH4YDU6MZWGV7H33IWEHUUIMPWPWOUVC7TMQUMI34KMORXM4IH3HHR3OHGFZJUIG"
[sudo] password for vboxuser:
--2026-05-01 12:31:03--  https://rmo86803.live.dynatrace.com/api/v1/deployment/installer/agent/unix/default/latest?arch=x86
Resolving rmo86803.live.dynatrace.com (rmo86803.live.dynatrace.com)... 34.247.97.7, 54.73.107.179, 34.250.81.163
Connecting to rmo86803.live.dynatrace.com (rmo86803.live.dynatrace.com)|34.247.97.7|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 289198887 (276M) [application/octet-stream]
Saving to: ‘Dynatrace-OneAgent-Linux-x86-1.335.58.20260423-145051.sh’

race-OneAgent-Linux-x8   6%[>                          ]  17.04M  5.36MB/s    eta 49s
```

```
vboxuser@virtualisation:~/login-web-app$ ls
app.py                                                    init.sql
docker-compose.yml                                        README.md
Dockerfile                                                requirements.txt
Dynatrace-OneAgent-Linux-x86-1.335.58.20260423-145051.sh  templates
vboxuser@virtualisation:~/login-web-app$
```

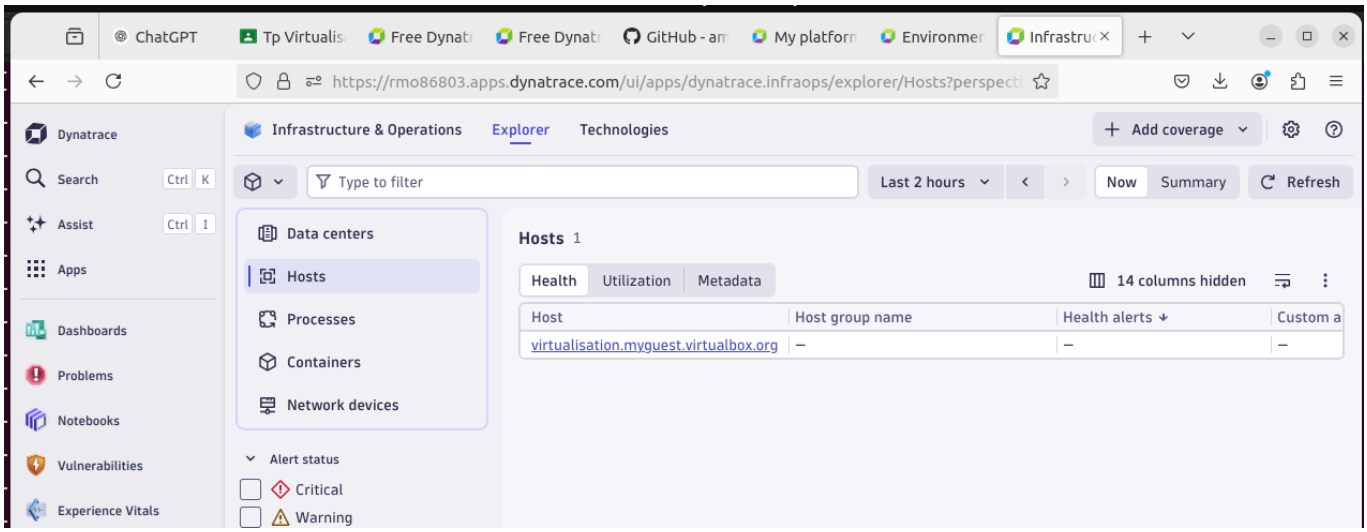
```
vboxuser@virtualisation:~/login-web-app$ sudo chmod +x Dynatrace-OneAgent-Linux-x86-1.335.
58.20260423-145051.sh
vboxuser@virtualisation:~/login-web-app$
```

Show Applications

```
vboxuser@virtualisation:~/login-web-app$ sudo ./Dynatrace-OneAgent-Linux-x86-1.335.58.2026
0423-145051.sh
12:34:03 Checking root privileges...
12:34:03 Process real user: root, real ID: 0
12:34:03 Process effective user: root, effective ID: 0
12:34:03 Process root access: true
12:34:03 Logging to /var/log/dynatrace/oneagent/installer/installation_54800.log
12:34:03 Installation started, version 1.335.58.20260423-145051, build date: 23.04.2026, P
ID 54800.
12:34:03 Detected platform: LINUX arch: X86_64
12:34:04 Checking /opt/dynatrace/oneagent for prohibited mount flags
12:34:04 Checking free space in /opt/dynatrace/oneagent
12:34:04 Extracting...
12:34:06 Unpacking. This may take a while...
12:34:37 Unpacking complete.
12:34:37 Checking if an operating system is supported...
12:34:37 The operating system is supported, continuing installation.
12:34:37 Moving new binaries into lib folders...
```

Étape 3 : Vérifier que l'agent "parle" à Dynatrace

```
vboxuser@virtualisation:~/login-web-app$ sudo systemctl status oneagent
● oneagent.service - Dynatrace OneAgent
     Loaded: loaded (/etc/systemd/system/oneagent.service; enabled; vendor preset: enable>
     Active: active (running) since Fri 2026-05-01 12:35:05 WAT; 1min 51s ago
    Process: 56321 ExecStart=/opt/dynatrace/oneagent/agent/initscripts/oneagent start (co>
   Main PID: 56398 (oneagentwatchdo)
      Tasks: 58 (limit: 2260)
     Memory: 314.5M
        CPU: 4.825s
     CGroup: /system.slice/oneagent.service
             ├─56398 /opt/dynatrace/oneagent/agent/lib64/oneagentwatchdog -bg -config=/op>
             ├─56406 oneagentos -Dcom.compuware.apm.WatchDogTimeout=900 -watchdog.restart>
             ├─56482 /opt/dynatrace/oneagent/agent/lib64/oneagenteventstracer --logdir /v>
             ├─56488 oneagentextensions -Dcom.compuware.apm.WatchDogTimeout=900 -Dcom.com>
             ├─56532 oneagentloganalytics -Dcom.compuware.apm.WatchDogTimeout=900 -Dcom.c>
             ├─56534 oneagentnetwork -Dcom.compuware.apm.WatchDogTimeout=900 -Dcom.compuw>
             ├─56539 /opt/dynatrace/oneagent/agent/lib64/oneagenthelper -p 51199 --contai>
             ├─56542 /opt/dynatrace/oneagent/agent/lib64/oneagenthelper -p 1023 --trace-p>
             └─56633 /opt/dynatrace/oneagent/agent/lib64/oneagentebpfdiscovery --log-dir >

May 01 12:35:05 virtualisation systemd[1]: Starting Dynatrace OneAgent...
May 01 12:35:05 virtualisation oneagent[56399]: 12:35:05 Dynatrace OneAgent service start>
May 01 12:35:05 virtualisation systemd[1]: Started Dynatrace OneAgent.
lines 1-22/22 (END)
```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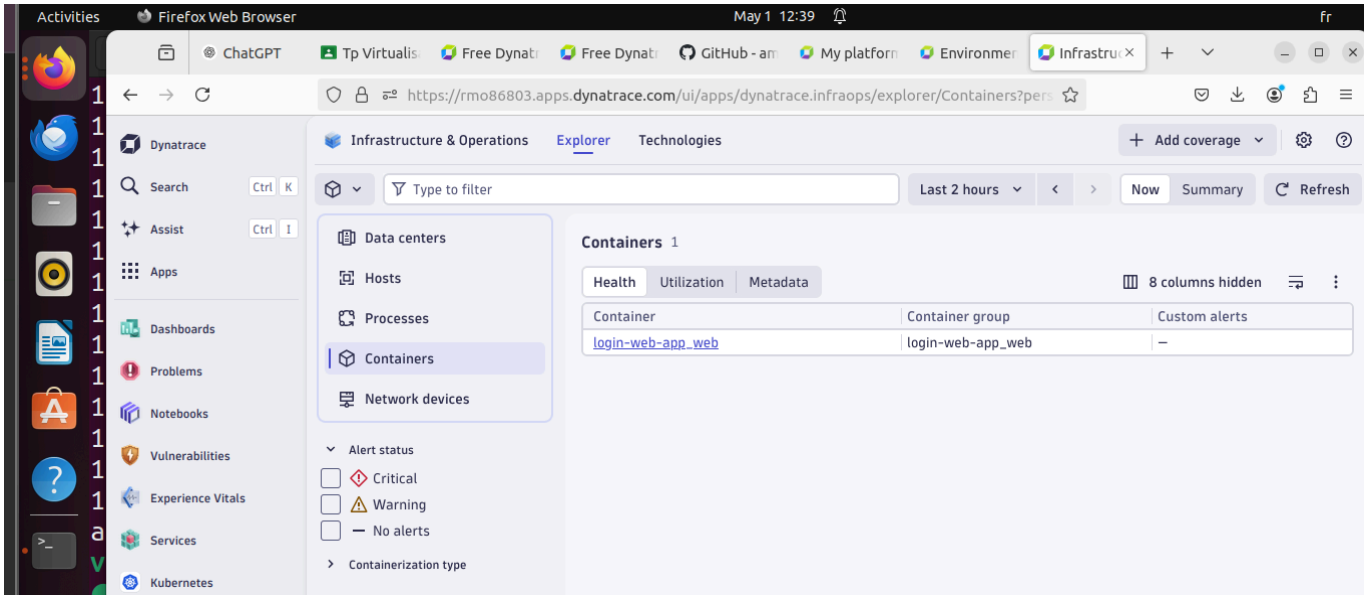
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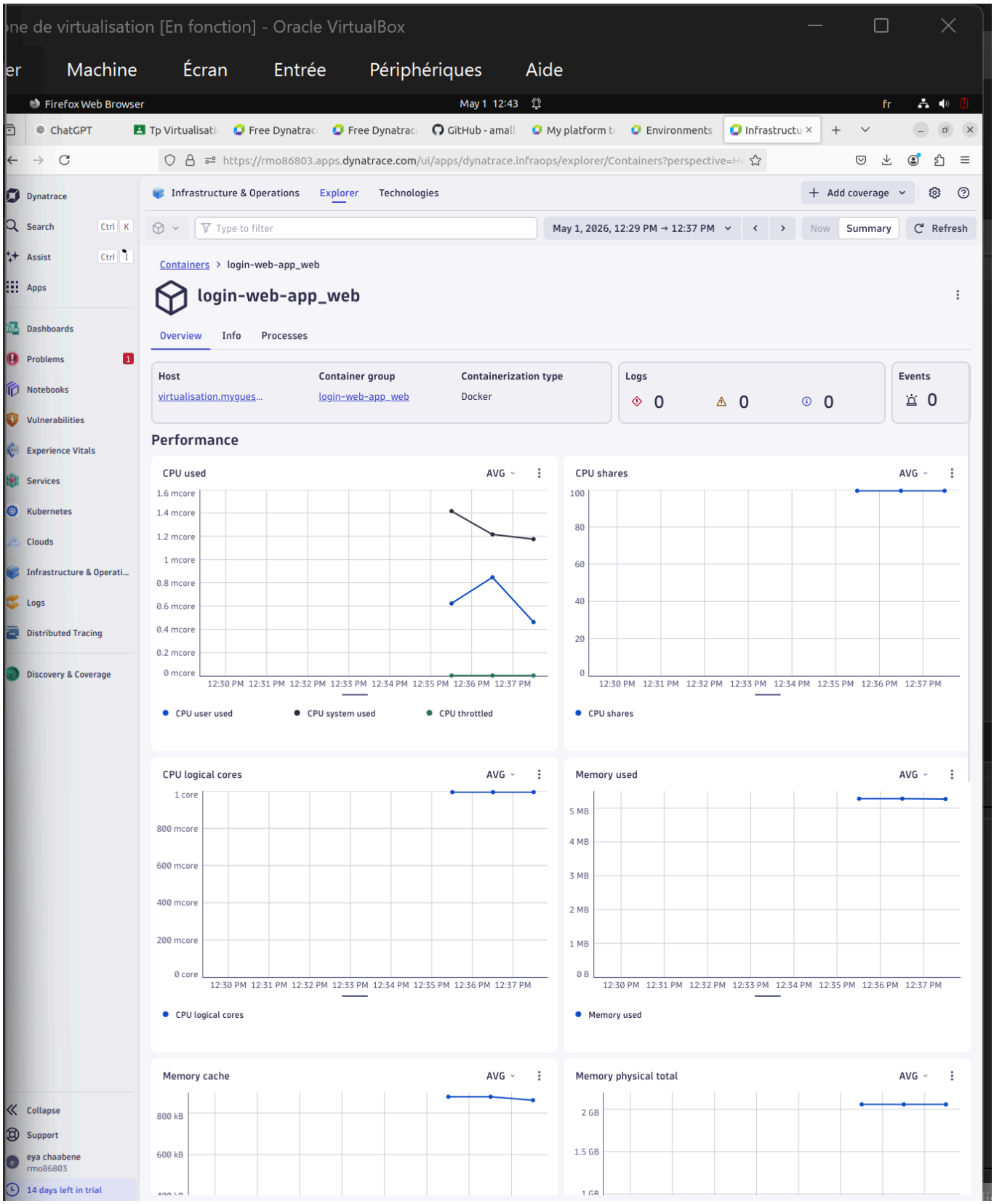
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Containers > login-web-app_web
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Overview
Info
Processes
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Container group
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Containerization type
Docker
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Events
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Performance
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1.2 mcore
1 mcore
0.8 mcore
0.6 mcore
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0.2 mcore
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CPU user used
CPU system used
CPU throttled
CPU shares
AVG
100
80
60
40
20
0
12:30 PM 12:31 PM 12:32 PM 12:33 PM 12:34 PM 12:35 PM 12:36 PM 12:37 PM
CPU shares
CPU logical cores
AVG
1 core
800 mcore
600 mcore
400 mcore
200 mcore
0 core
12:30 PM 12:31 PM 12:32 PM 12:33 PM 12:34 PM 12:35 PM 12:36 PM 12:37 PM
CPU logical cores
Memory used
AVG
5 MB
4 MB
3 MB
2 MB
1 MB
0 B
12:30 PM 12:31 PM 12:32 PM 12:33 PM 12:34 PM 12:35 PM 12:36 PM 12:37 PM
Memory used
Memory cache
AVG
800 kB
600 kB
Memory physical total
AVG
2 GB
1.5 GB